Christopher Suarez

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SUMMARY

Robotics engineer with 10 years experience focused on development & deployment of autonomous systems

EDUCATION

M.S. Mechanical Engineering, University of Texas at Austin 2019

B.S. Mechanical Engineering, University of Florida 2014

EMPLOYMENT

Apptronik Austin, TX

Operations Technology Lead Feb. 2026 – Present

- Created OT team, managed tasking through Jira and led OT hiring efforts
- Automated critical fleet-wide deployment and diagnostic processes via Ansible & crontab on server
- Supported and improved internal fleet monitoring through Zabbix
- Implemented harbor docker registry mirror to mitigate local network bandwidth issues across fleet

Senior Software Support Engineer July 2025 – Feb. 2026

- Primary software engineer responding to software / firmware issues with deployed robots
- Performed all OTA software upgrades to the fleet and PR/MR'd various OTA software improvements
- Supported critical robot deployment and demo events domestically and internationally
- Created diagnostic tools for Field Service and Operations teams for ECAT and firmware diagnosis

TORC Robotics (contractor through Actalent) Austin, TX

Software Integration Engineer Nov. 2024 – Present

- Streamlined Jenkins CI/CD pipeline using Bazel, reducing CI build times by 33 – 50%
- Created CPU & Memory resource usage tool to assess sensor driver performance at runtime and in CI
- Updated IMU and Lidar sensor drivers to modern internal fault framework
- Debugged hardware and sensor firmware issues for Hardware-in-the-loop (HIL) Lab
- Expanded sensor validators to new sensor drivers ensuring performance to meet requirements

UT Austin Nuclear & Applied Robotics Group Austin, TX

Research Engineering Scientist Associate IV Sept. 2019 – Nov. 2024

Summary

- Oversaw and performed over 100 technical demos directly leading to >$10M in project funding to UT
- Expertise in setting up, debugging, and operating robotics systems with ROS
- Managed development teams for multiple research integration projects
- Mentored new engineer and student hires across varying research efforts

Army Futures Command UTDD Project SPEED

- Lead demo engineer overseeing a large-scale on-site technical demonstration with a team of 12
- Stood up a new applied research lab focusing on large vehicle outdoor autonomy
- Integrated Robot Technology Kernel (RTK) 22 onto a previously incompatible Hunter WOLF platform
- Created a simulation test environment in Unreal Engine

Augmented Robot Environment (AugRE) Project

- Developed and dockerized software stack to co-localize robots with Hololens2 AR Headsets
- Managed dev team creating software for AR in Unreal Engine and robotics in ROS/C++
- Setup and managed LAN & WAN network for multiple autonomous and human agent comms
- Performed dozens of live hands-on demonstrations to VIPs, Gov Officials, Army Generals, CEOs

Army Futures Command UTDD Integrated Prototype Project

- Contributed features, sensor integrations, and critical fixes to Army Research Labs' autonomy stack
- Integrated ARL Phoenix navigation stack with custom research platform hardware
- Conducted multi-week study to quantify long-term autonomy viability of various UT research efforts
- Created custom robot and environment assets for Unity environment for simulation development
- Developed object persistent localization software fusing camera, LIDAR, and mobile robot data

AI2C ATR-MCAS Project

- Stood up an Army research platform for autonomous navigation using GPS, IMU, LIDAR, Segway base
- Incorporated capability to visualize our autonomous software stack with ATAK (Android Tactical Awareness Kit) via Cursor-On-Target (CoT) XML messaging
- Extended ATAK visualization capability to our AR AugRE system allowing human agents to appear on ATAK and interface with autonomous agents without co-localizing directly with AR anchors
- Performed yearly deliverable hands-on live demonstrations on-site for Army VIPs

SRNL H-Canyon Tunnel Inspection Project

- Developed 3D obstacle detection software for an off-road dual arm mobile manipulator via ROS / C++
- Designed and constructed a 9'x12'x36' above ground tunnel as a mockup harsh test environment
- Performed final semi-autonomous demonstration of tunnel inspection inside a mock environment

Lockheed Martin Missiles & Fire Control — Orlando, FL
Systems Engineer — June 2014 – July 2016

- Created system level engineering requirements and test procedures
- Conducted program critical tests in-plant and on-site - domestically and internationally
- Presented Test Readiness Reviews to off-program engineers for milestones for numerous programs
- Designed GUIs to operate and test complex machinery and contributed to the end product UI design

PERSONAL PROJECT

Baseball Simpleton — Apr. 2026

- Analyzed MLB data from statcast era into accessible UI (Typescript / React) for player analysis
- Split data into year-by-year SQL databases and one meta SQL database to allow for fast queries
- Implemented deployment through AWS EC2 system, DBs on S3 bucket, and crontab to update nightly

PUBLICATIONS & PRESENTATIONS

IROS 2023 XR-ROB Workshop — Oct. 2023
Abstract/Presentation – *AR UI for Command, Control, and Supervision of Large Multi-Agent Teams*

IEEE Internation Conference on Robot & Human Interactive Communication (RO-MAN) 2022 — Sept. 2022
Paper - *AugRE: Augmented Robot Environment to Facilitate Human-Robot Teaming and Communication*

Waste Management Symposia 2019 — Mar. 2019
Paper – *Savannah River Site H-Canyon Tunnel Inspection LiDAR Mapping Solution*
Paper – *Savannah River Site H-Canyon Advanced Technology Demonstration (ATD) Program Review*

SKILLS

- Programming Languages: Python, C++, Typescript, Shell
- AI Tools: Claude Code, OpenAI Codex, Gemini,
- Dev Tools: Git, Docker, Ansible, Zabbix, Bazel, Jenkins, AWS EC2 & S3, Jira
- Simulation: Unreal Engine, Gazebo, Unity
- Software: ROS, Solidworks, Visual Studio, MatLAB, LabVIEW, QT, Linux, MacOS
- IT: LAN / WAN networking, embedded machines, edge compute
- Hardware: Additive manufacturing, lathes, mills, CNC…
- Systems: Requirements, validation & testing, technical reporting, multi-disciplinary collaboration